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Flood-affected homes in England lose value and are owned longer than those in unaffected areas



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Flooding is one of the costliest weather-related disasters. In a warmer, wetter world, future flood events are expected to become more frequent and severe. Previous research shows that flooding devalues property in England, but the effect on length of home ownership is unexplored. We analysed 27 million residential property sales over 28 years in England to examine how flood zoning, nearby flooding, or inundation affected ownership length and property values. We find that flooded properties typically lose 3% of their value immediately, rising to a 10% reduction after 15 years. This aggregate £5.6 billion loss persists because floods are not forgotten, suggesting the market has moved beyond amnesia-driven rebound effects towards sustained repricing of flood risk. Moreover, flood-affected properties are owned on average for a decade longer than those in unaffected areas, especially among lower-priced homes. Our results have implications beyond England for housing markets, mortgage lending and insurers.

Flooding is a major global concern due to the significant economic costs and impacts on communities^{1–4}. In England, one in six properties (5.7 million) are currently at risk of flooding, of which 2.8 million are exposed to sea or river flooding⁵. In autumn 2023, Storms Babet, Ciaran, and Debi caused total insurance payouts exceeding £550 million, whilst in winter 2024 Storm Henk brought widespread flooding that affected over 2200 properties^{6,7}.

Post-flood, inundated homes are discounted in price – especially properties in lower price quartiles^{8,9}. Similar patterns have been reported for house prices in Australia, Germany and the United States, perhaps signalling growing awareness of flood risks among buyers and investors globally^{4,8,10,11}. Moreover, properties with a history of flooding or those in flood-prone areas appreciate in value at slower rates than those in lower risk locations^{8,9,12,13}.

Such trends have implications for home insurance and mortgage availability in areas at high risk of flooding^{14–17}. Major UK banks already regulate loan amounts for high-risk properties, declining some applications due to intolerable flood risk¹⁸. Meanwhile, the Bank of England warns of borrowing difficulties for flood affected homeowners¹⁹. This raises major policy implications for public and private institutions concerning the saleability and insurability of properties at risk of, or historically affected by flooding—underscoring the need to maintain affordable flood insurance^{20–23}. Nonetheless, over the last decade, 1 in 13 new homes were still built in Environment Agency (EA) Flood Zone 3 (FZ3) (comparable to the US Special Flood Hazard Area), which has at least 1% annual probability of

river flooding^{24,25}. Moreover, no homes built since 2008 are covered by the national reinsurance scheme, Flood Re, which is due to expire in 2039.

These issues raise concerns for homeowners about the financial impact of flooding on the value, insurability, and saleability of their property^{26,27}. Home ownership is deeply ingrained in the English psyche, symbolising both a personal milestone and a means of capital accumulation^{28,29}. Real estate also forms a large portion of individual wealth, hence changes to property values affect financial status and broader financial systems^{4,30,31}. Our study significantly extends a dataset previously used to investigate links between floods and property prices in England^{9,13,26,32} and, for the first time, investigates the impact of floods on length of home ownership.

Public availability of flood zone maps, historical flood extents (Fig. 1), and residential property transaction records since 1995 makes England an appealing case study for investigation. Moreover, with home ownership rates (63% in 2021) broadly comparable to those in France (65%), Germany (49%) and the US (65%), England may be indicative of trends in other property markets in the global north^{33–35}.

This study evaluates the overarching impact of flooding—including inland fluvial and sewer flooding and coastal flooding—on length of home ownership and property prices in England using modelled flood exposure and inundation history. We then consider the wider economic implications for homeowners, insurers, mortgage lenders and public institutions. Finally, we suggest future research priorities to enhance understanding and management of these impacts.

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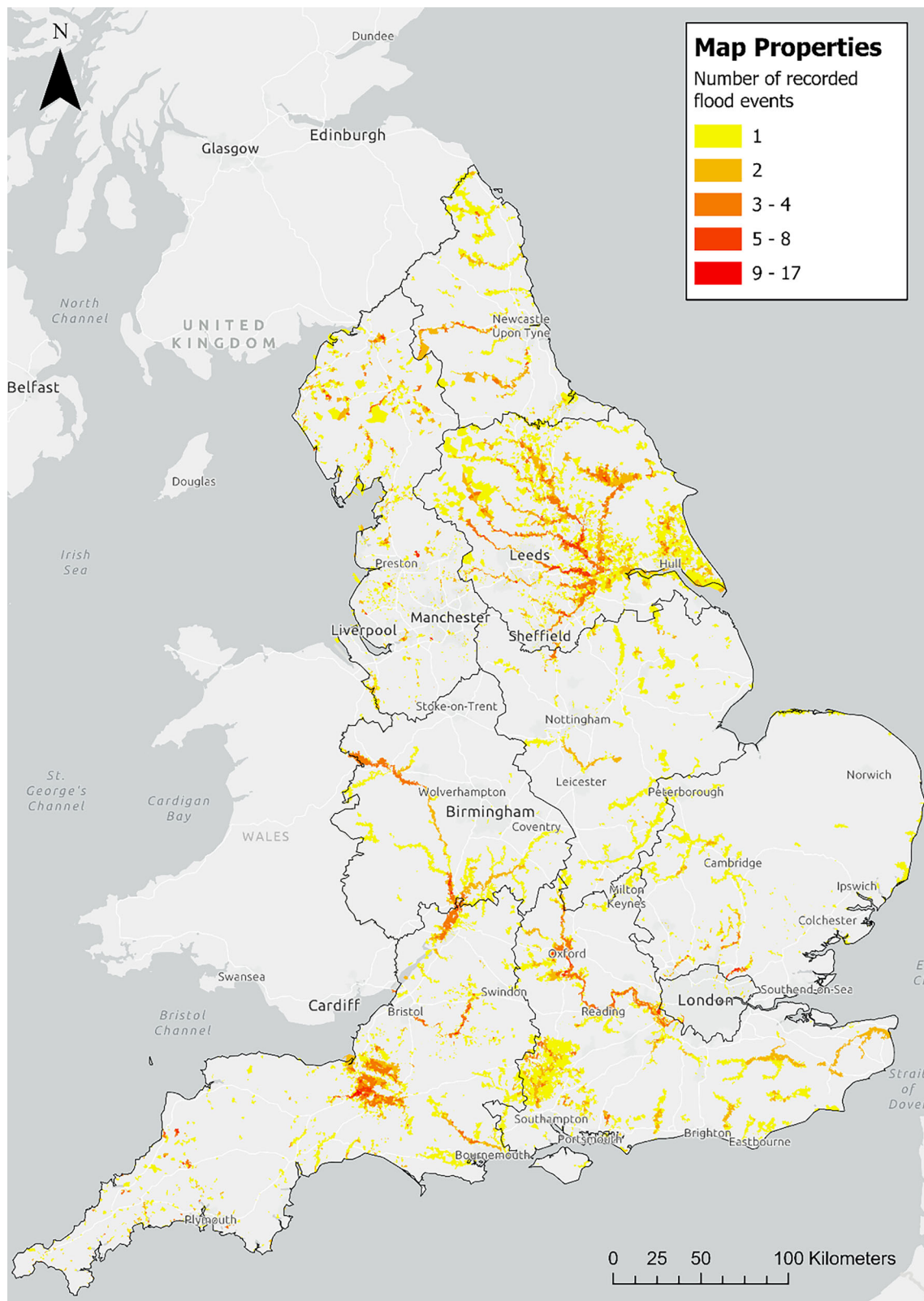


Fig. 1 | Number of recorded floods by postcode district in England between 1st January 1995 and 1st January 2023. In total, there were 78,025 events covering an area of 2,723 km². Fluvial flood events account for 89% (2,414 km²), whereas coastal and sewer flood events covered 6% (170 km²) and 5% (139 km²), respectively⁹³.

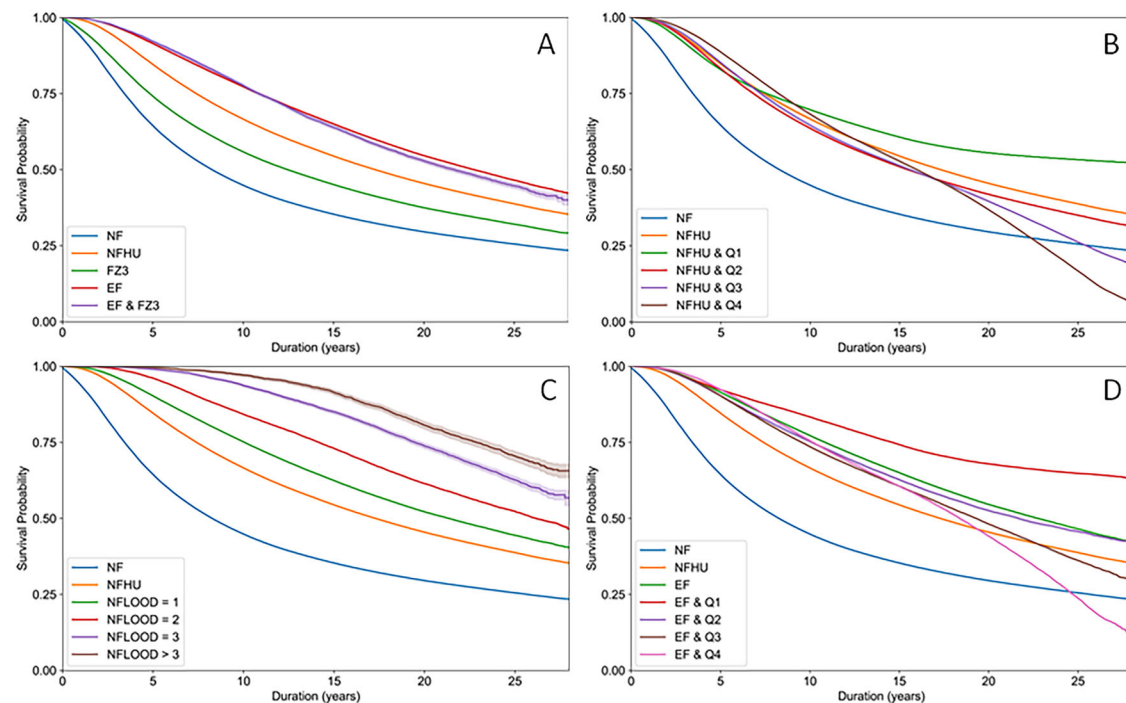
Impact of flooding on home ownership duration

We analysed 27 million residential property transactions in England spanning a 28-year period (1st January 1995–1st January 2023), to detect the impact of historical flooding and flood risk zonation (Table 1) on home ownership duration (Fig. 2). We discover that nationally, the median length

of home ownership was 4 years in areas unexposed to flooding, rising to 22 years in the worst flood-affected areas (Table 2). In comparison, the median length of home ownership was 15 years for places that experienced at least one flood event.

Table 1 | Classification of properties into sub-populations based on history of flooding and/or exposure to flood risk

Condition	Description
No Flooding (NF)	No flood was recorded within the property district. In England, a district is a level of subnational division used for the purposes of local government. We use district boundaries that were correct as of 31st December 2022.
Nearby Flooding, Home Unaffected (NFHU)	A flood event was recorded within the district (average land area of 504 km ²), but not within the property postcode-district. For context, postcode-districts on average have 15 properties ⁶⁰ .
High Flood Risk Area (HFRA)	The property is in Environment Agency (EA) Flood Zone 3 (FZ3). This applies to areas with at least 1 in 100 annual probability of flooding by rivers or the sea.
Experienced Flooding (EF)	At least one flood was recorded within the property postcode district.
Experienced Flooding, High Risk Area (EFHRA)	There was at least one flood within the postcode district, and the property is in EA FZ3.

**Fig. 2 | Kaplan-Meier survival curves, showing how long (years) sub-populations exposed to flooding remained in their property, with higher curves indicating longer residence durations. Durations are shown for A key spatially defined sub-**

populations, **B** the effects of nearby flooding but home unaffected, across different price quartiles, **C** different flood frequencies and **D** homes that experienced flooding, stratified by price quartile. Sub-population definitions are in Table 1.

There are various explanations for this prolongation. It may reflect homeowners avoiding short-term financial losses by waiting for a recovery in property values post-flooding; the time needed to process insurance claims and/or complete repairs; place attachment; amenity value associated with proximity to water bodies; lack of alternative housing; and/or outright unsalability of homes^{36–39}.

Home ownership duration matters because it drives the volume of activity in housing markets, determines the speed of adjustments and completeness of price revelation, and is a central metric when calculating turnover rates^{40,41}. Therefore, home ownership durations can be indicative of underlying liquidity in housing markets – with longer home ownership suggesting less liquidity⁴¹.

Flooding and flood risks extend the time required to sell properties in affected areas, mirroring the effects observed during economic downturns, namely, house price volatility, reduced sales volume, and increased illiquidity^{42,43}. This prolongation stems from the seller's challenge in attracting a broad pool of potential buyers resulting in a mismatch between seller price expectations (often with a pre-flood reference point) and buyer hesitancy to accept associated flood risks^{42,44–48}.

However, the impacts of flooding—typically analysed in the form of price discounts for affected properties—are generally understood to fade

with time. This concept is widely described as ‘flood amnesia’. The expectation is of an initial downturn in prices within flood-impacted areas, leading to a slowdown in sales and consequently longer home ownership because of repairs, and time spent on the market^{49,50}. However, as time goes on, prices are expected to recover as memories of the event fade, and consequently, sale volumes and market liquidity improve^{8,9,14,50,51}.

Contrary to the above, our findings (Fig. 2, Table 2) show that flooding has a marked and enduring impact on home ownership length. This effect is observable not just in frequently flooded areas but also in their vicinities. This first line of evidence challenges the notion of flood amnesia, suggesting we may have reached the tipping point Pryce et al envisaged—where once myopic and amnesiac views give way to a sustained repricing of flood risk¹⁴.

Declining housing turnover in flood-affected areas can impact property-tied sectors, and wider economic activity^{52,53}. Housing market turnover and the total amount borrowed to purchase residential property are closely related, indicating that as turnover decreases, so too does the amount of housing credit that is extended⁴¹. Such a contraction has consequences for mortgage lenders in terms of housing credit and revenue growth: as loan originations decline, so does profitability. Given the connectedness of turnover rates and property prices, declines in the former can

Table 2 | Home ownership durations (years) from the Kaplan-Meier survival curves (Fig. 2)

Condition	25th Percentile (years)	Median (years)	75th Percentile (years)	Mean (years)	% Sold	% Unsold	Total
NF	1.9	4.3	8.4	6.2	46.1%	53.9%	12,793,591
NFHU	5.5	10.6	17.5	11.9	45.7%	54.3%	13,608,381
NFHU & Q1	5.5	12.8	21.7	13.6	40.0%	60.0%	4,012,904
NFHU & Q2	5.3	10.0	16.5	11.3	48.1%	51.9%	3,502,212
NFHU & Q3	5.6	10.2	16.3	11.1	48.6%	51.4%	3,234,415
NFHU & Q4	5.8	10.2	16.1	11.1	47.7%	52.3%	2,858,850
FZ3	3.3	7.1	14.6	9.1	48.1%	51.9%	665,738
EF	8.6	15.2	20.4	14.6	41.8%	58.2%	303,321
EF & Q1	10.3	19.0	23.7	17.0	30.8%	69.2%	86,003
EF & Q2	7.9	14.3	19.4	14.0	42.2%	57.8%	69,504
EF & Q3	7.9	13.9	18.6	13.6	46.3%	53.8%	68,893
EF & Q4	8.7	13.9	18.1	13.5	49.5%	50.5%	78,906
EF & FZ3	9.5	15.6	19.8	14.8	43.9%	56.1%	20,719
NFLOOD = 1	8.1	14.4	19.6	14.1	43.2%	56.8%	251,628
NFLOOD = 2	11.2	17.1	22.0	16.6	38.0%	62.0%	37,008
NFLOOD = 3	15.2	19.8	23.7	18.9	29.1%	70.9%	10,388
NFLOOD > 3	17.5	22.3	25.1	20.8	24.6%	75.4%	4,297

'% Sold' shows the percentage of homeowners in each sub-population observed to sell their property. '% Unsold' is the percentage of homeowners who remained in the same property beyond the observation period. 'Total' is the overall number of sales in each sub-population. Definitions of each condition are given in Table 1.

precipitate falling property prices which could present additional risks for mortgage lenders coupled with tightening liquidity^{54–56}.

These dynamics could drive a challenging cycle in high flood-risk areas, evidenced by mortgage lenders rationing services due to declining turnover, shrinking housing credit, reduced insurance coverage, and elevated financial risks^{57–60}. This rationing, characterised by higher mortgage rates and insurance premiums to account for increased risk, further inhibits housing market activity, potentially leaving homeowners in flood-affected areas effectively 'locked-in'. Although public interventions such as the National Flood Insurance Programme (US) and Flood Re scheme (UK) aim to mitigate such effects by offering more affordable insurance options, observations of longer ownership lengths suggest that such schemes may fall short in revitalising and supporting market activity in flood-affected areas.

Impact of flooding on property prices

Using a repeat-sales hedonic price model, we examined 12 million pairs of sales—instances where a property was sold at least twice, between 1st January 1995 and 1st January 2023. We find properties that experienced at least one flood event (EF) lost on average 2.9% ($\pm 0.7\%$) of their value (see supplementary material for full results). This price change is measured relative to what would otherwise have occurred for non-flooded properties. Among EF properties, the prices of flats and semi-detached properties were discounted on average by 6.2% ($\pm 1.2\%$) and 1.3% ($\pm 1.2\%$), whereas detached properties increased in price by 0.5% ($\pm 1.2\%$). These changes indicate the differential effect of flooding on the price of different types of properties relative to non-flooded terraced properties. NFHU properties were discounted on average by 10.7% ($\pm 0.1\%$). Among these properties, detached and flat properties were most severely impacted, experiencing an average price reduction of 20.4% ($\pm 0.2\%$), and 20.9% ($\pm 0.2\%$), respectively, compared to semi-detached properties (12.5% $\pm 0.1\%$). These price changes are relative to non-flooded terraced properties.

The impact of flooding on property prices is commonly understood to exhibit amnesiac and myopic traits^{8,9,14,50}. Prior to a flood, property prices may reflect near to zero-flood risk as market participants either disregard or are unaware of the inherent flood risks. Immediately after a flood, the realisation of flood risk prompts a swift reassessment, typically resulting in a significant reduction in property values to reflect the newfound flood risk

awareness. Over time, the memory of the flood event fades – driven by a combination of memory decay and turnover of residents who will be less familiar with an area's flood history. Hence, property prices are expected to recover to levels reflecting diminished flood risk perception.

Contrary to this expectation, we find that property prices post-flooding do not exhibit such traits. Instead, we uncover a more persistent awareness of flood risks, particularly in areas with known histories of flooding. Properties that experienced flooding (EF sample) showed smaller price discounts than those near flood events yet unaffected directly (NFHU sample).

Properties in areas with a known flood history may have such risks more accurately reflected in their prices, leading to less dramatic post-flood price adjustments. In these locales, the risk of flooding is a known quantity, effectively 'baked into' the price, leading to market behaviours that are perhaps more predictable^{8,61}. Adjustments here are less about responding to shock and more about recalibrating prices based on more recent information, indicating a different framing of risk that incorporates past flood events as a reference point for future expectations^{45,62,63}. If flood amnesia were prevalent, and housing markets forgot about the severity of flooding, we would expect to see larger discounts following flood events as markets react sharply to the 'surprise' of the event. However, the observed differences in discounts between EF and NFHU samples challenge the traditional model of amnesiac and myopic behaviours typically expected in flood-impacted housing markets. Our findings related to homeownership length further reinforce the view that housing markets may have progressed beyond initial amnesia-driven rebound effects towards a sustained repricing of flood risk and permanent erosion of value for flooded properties¹⁴. Longer homeownership durations in EF areas contribute to, and reinforce, community-wide awareness and understanding of flood risks⁶⁴, significantly reducing the risk of memory decay due to low turnover.

Flood amnesia may have been more prevalent previously, but growing public visibility of climate change has heightened vigilance towards flood risks^{65,66}. In areas with a documented flood history, this greater awareness is likely fostering less severe price shocks. Consequently, market agents are updating the reference point from which flood risk is assessed to reflect the true environmental risks more accurately in their valuations^{48,67–69}. Meanwhile, Flood Re may have helped stabilise flood-prone property markets by

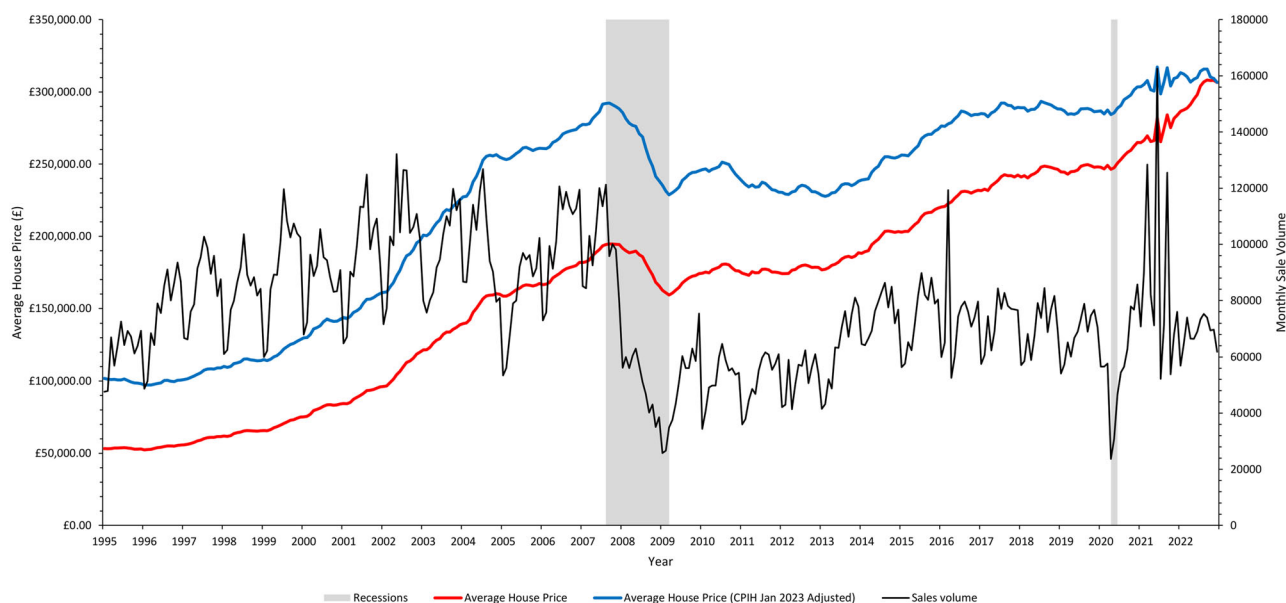


Fig. 3 | Monthly average house prices and transaction volumes in England, 1st January 1995 to 1st January 2023¹⁰⁰. The red line shows the nominal Land Registry average price for all property types. The blue line is the same series adjusted to January 2023 prices using the CPIH index¹⁰¹. The black line (right-hand axis) plots

total monthly sales volumes. Shaded bands mark UK recessions. Over the three defined time horizons, nominal house prices increased by 212% (1995–2008), 35% (2009–2015), and 39% (post-2016), while CPIH-adjusted prices rose by 134%, 16% and 11%, respectively (see supplementary material).

ensuring continued access to affordable insurance, particularly for older homes. However, its limited visibility during transactions and behind-the-scenes operation mean its effect on buyer behaviour and pricing is likely indirect and not easily detected in market data.

Notably, steeper price discounts for properties in NFHU areas, could be attributed to flooding impacting previously unaffected areas or more frequent nearby events. Properties in areas without a history of flooding are likely to undergo a sudden devaluation as the market rapidly reassesses the potential for future flooding based on this new and unexpected information^{50,68}. This recalibration may be driven by an update in beliefs around the likelihood of future flooding, which originated from a reference point assuming minimal or no flood risk^{68,69}. Price adjustments are not solely a reaction to the presence of nearby flood waters, but also to the unexpected nature of the event, altering the perceived security of investments previously considered safe. Areas with no previous flood history may serve as a baseline against which any new flood risks are evaluated, emphasising the impact of sudden shifts in perceived safety^{45,62,63}. Recent studies indicate signs of overvaluation in housing markets exposed to flood risks, with prices and sales volumes in gradual decline^{4,48,69}. As market awareness of flood risks grows, incorporating both present and anticipated future flooding, we may observe the deflation of a real estate bubble, correcting for previously underestimated flood risks¹⁴.

Longer home ownership for properties that experienced flooding may mask the true financial impacts on property prices, as homeowners delay selling in the hope of a market recovery. Notably, among properties subjected to more than three flood events, 75% of homeowners remained in their properties beyond the study period (Table 2). This highlights a considerable, yet unaccounted for, financial toll and ongoing hardship for residents in frequently flooded areas—meaning that the true price discount due to flooding remains unrealised in market valuations until the point of sale.

Price discounts for flats in the aftermath of flooding highlight their unique vulnerabilities compared to terraced, detached, or semi-detached properties. Flat owners, especially in larger complexes, must navigate repair coordination and can face additional difficulties if communal infrastructure is damaged⁷⁰. Moreover, differentiating impacts of flooding on sales data between floors is challenging, since ground-level units are particularly vulnerable. Discounting may also be influenced by fire safety concerns

following the Grenfell Tower fire in June 2017, where cladding issues have reduced flat valuations and hindered sales and mortgage approvals^{71–74}.

Our findings also reveal disparities across segments of the housing market, particularly when comparing the lowest (Q1) and highest (Q4) price quartiles in NFHU and EF areas. These disparities are likely caused by a combination of fixed and proportional disutility components and cost scaling relating to property size for lower price quartile properties, resulting in a steeper discount in price. Flood events impose a fixed disutility (e.g., cleanup, repairs) that applies regardless of property value, alongside a proportional component that scales with property size. Flood damage will likely affect a greater proportion of usable space in smaller homes (correlated with lower price homes), rendering a larger share uninhabitable or damaged, amplifying the disutility for lower-value properties. Properties in Q1, not only face steeper price discounts post-flooding, but also experience longer periods of home ownership. One explanation could be the compounded difficulties faced by these homeowners, such as higher financial burden of flood damage relative to property value. Moreover, Q1 homeowners may be more likely to forgo flood insurance due to cost constraints^{75–77}.

Period and cohort effects

To isolate the influence of macroeconomic and institutional conditions on housing returns, we include period dummy variables, defined by the calendar window in which the second sale of each sale-pairing occurs⁷⁸. The reference period is 1995–2008, a time of sustained real house price growth and relatively subdued CPIH inflation (see Fig. 3). Because all sale prices have been adjusted to January 2023 prices, the estimated coefficients reflect inflation-adjusted price differentials across resale periods.

The period effects are both large and consistent. Whether the second sale falls in the 2009–2015 (−32.6%) or in the post-2016 window (−33.5%), the price ratio shrinks to about two-thirds of what it was in the reference window—meaning properties sold in the latter two periods saw price changes around a third lower relative to properties sold in the first period on average. The size and persistence of this penalty are consistent with the macroeconomic narrative of the period. The global financial crisis removed about one-fifth from real house values in 2008–2009 and ushered in a long period of austerity, tight credit controls and sluggish wage growth, dampening any real price growth for property prices^{79,80}. Meanwhile, post-2016

coincided with Brexit-related uncertainty and, more recently, a post-pandemic surge in CPIH that has eroded much of the nominal house price boom of 2021–2022⁸⁰ (see Fig. 3).

By contrast, the cohort indicators—defined by the calendar year of a property's first recorded flood—exert only a modest influence once age and period effects are controlled (see supplementary material). Properties first flooded before 2009 carry a real price change just two percentage points below never-flooded comparators (−2.1%), whereas those first flooded in 2009–2015 (−0.5%) or after 2016 (−1.2%) sit one to one-and-a-half percentage points lower still. The small magnitudes suggest that the insurance regime prevailing at the moment a house first enters the at-risk pool matters far less than the broader macro environment during which the resale occurs. In other words, the steepening of the price-flood-frequency curve documented by Pryce et al appears to have operated mainly through a system-wide shift—insurers tightening cover, lenders adjusting valuations—rather than through permanent scarring of particular flood cohorts¹⁴.

Taken together, our results suggest a two-layer adjustment mechanism. Age effects govern the shape of the post-flood price path, whereby property prices recover in the first few years, likely driven by insurance-funded repairs, then progressively lose ground, likely due to growing awareness of flood risks. Period effects determine the average level of that path, lowering real appreciation by about one-third across the entire housing market after 2008. Cohort effects play only a secondary role once these other channels are recognised. The evidence, therefore, favours a behavioural-institutional reading of the market—the delayed and then accelerating price penalty reflects an interaction between boundedly rational buyers, evolving risk information, and the changing stance of insurers and lenders, rather than a purely forward-looking, perfectly informed arbitrage process.

Long-term property price recovery

Our analysis uncovers a concerning pattern in the aftermath of flooding: an immediate depreciation in property value as expected^{8,9,81,82}, but then the discount deepens to ~10% after 15 years, which is not recovered (Fig. 4). This pattern is consistent across all price quartiles, including higher-valued properties. Such sustained devaluation challenges the assumption of 'flood amnesia', whereby market adjustments are expected to be short-lived as public awareness of flood risks fades over time. Instead, we find enduring economic effects consistent with climate gentrification, suggesting markets have reached a tipping point with the strategic retreat of capital from areas perceived as high-risk^{12–14}.

When accounting for the negative impact of flooding—reflected in the flood-related price discounts across each recovery period—the aggregate property value (analysed in the EF sample) is estimated at £34.8 billion (see supplementary material). This represents a reduction in value of around

£5.6 billion ($\pm$ £364 million) due to flooding, averaging a loss of £43,347 per property. Although our findings quantify persistent impacts of flooding on property values, the true financial toll is likely to be much higher, as homeowners in flood-affected areas have held onto their properties beyond the study period—likely masking deeper price discounts.

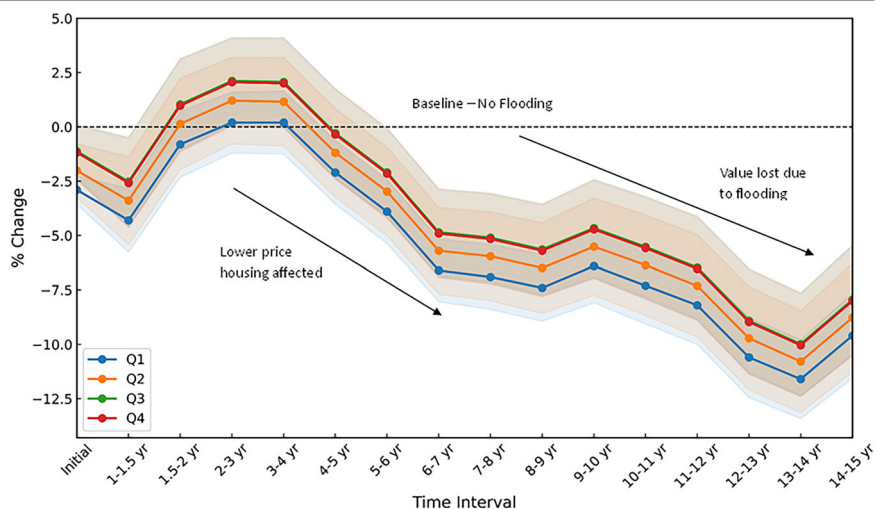
A study in England found that property prices rebound to pre-flood levels within 6–7 years following a flood event, supporting the flood amnesia paradigm⁹. However, their observation period spanned January 1995 to July 2014 (19.5-year period), compared to our end date of January 2023 (28-year period). Within the shorter dataset it is only possible to observe property sales occurring six to seven years post-flood for flood events up to July 2008. This cutoff point places a large portion of their observed sales of this type (63.7% of our data) during the economic recovery following the 2008/09 financial crisis, when property prices increased sharply year-on-year⁸³. This period's strong housing market could introduce systematic bias, inadvertently sampling a special and temporary set of circumstances when flood amnesia did occur.

Our longer study period benefits from covering diverse economic conditions. During this time, significant regulatory and informational changes have reshaped how properties re-enter the market. In May 2013, the UK Law Society issued flood risk management guidance for solicitors, reinforced by updates to the Property Information Form (TA6) that mandate flood impact disclosures for all residential transactions⁸⁴. Additionally, the public release of EA flood maps in December 2014 enhanced transparency about property-level flood risks⁸⁵. For further information on key policy changes during our study period, see the supplementary material.

Sustained devaluation in flood-impacted areas, that are already exhibiting reduced housing turnover and constrained liquidity, amplifies risks for mortgage lenders^{86,87}. Persistent negative equity increases the risk of homeowner default, which in turn inflates lender credit risk exposure, potential losses and capital requirements for bad debts, and thereby reducing profitability⁸⁷. If liquidity is already constrained and parts of a lender's mortgage portfolio are falling into negative equity, this limits their ability to offload high-risk assets quickly and raises the uncertainty of the realisable value of the collateral. This interplay of reduced liquidity and increased negative equity can create a dangerous cycle, where constrained asset liquidation and escalating default risks feed into each other, further destabilising the housing market and amplifying financial strain on lenders^{86–89}.

Taken together, our findings imply that flood risk in England now operates less as a temporary shock and more as a persistent, structural drag on property values and therefore household wealth in at-risk areas. Rather than prices fully recovering after an event, we observe a gradual deepening of the price discount over more than a decade, consistent with a slow revaluation of exposed assets as information, insurance and lending regimes

Fig. 4 | Changes (%) in property prices following a flood, by price quartiles Q1 to Q4 (inflation adjusted to January 2023). Shading denotes two standard errors around the central estimate. These data reveal initial and persistent impacts of floods on property prices, highlighting the greatest impact on the lowest priced (Q1) homes, especially more than 5 years after a flood. The black dashed line represents the relative price of a non-flooded Q1 terraced property.



evolve. Approaches that focus on short-run price effects or assume eventual convergence of prices in high- and low-risk areas are likely to understate the long-run impact of flooding on property values and on the equity that households hold in their homes. This may lead lenders, insurers and regulators to underestimate the scale and timing of climate-related risks building up on their balance sheets.

At the same time, the combination of prolonged homeownership durations and sustained devaluation in flooded and high-risk neighbourhoods, particularly in the lowest price quartile, indicates that flooding can restrict mobility for more vulnerable homeowners. Climate risks therefore enter the housing system not only by eroding collateral values, but also by thinning market liquidity and restricting exit options from high-risk locations. For regulators, lenders and policymakers, this underscores the need to treat flood risk as both a price and a liquidity channel, and to anticipate that institutional and informational reforms (including mapping, disclosure requirements and insurance arrangements) will shape when, where and for whom these long-run losses are realised^{23,88}.

Conclusions

Our findings uncover enduring impacts of flooding on property prices and length of home ownership in England. We show that flood-affected properties face significantly extended home ownership durations—up to a decade longer than those in unaffected areas. Homeowners in these areas are not only challenged by immediate depreciation in property values but also by obstacles to financial recovery and geographical mobility due to the stigma and reduced desirability of flood-prone properties. The risk of sustained devaluation in flood-affected areas can increase financial instability, with the potential for contagion effects across economic sectors tied to real estate. Additionally, the expected impacts of climate change, including more frequent and severe flooding, could exacerbate these challenges, leading to even greater economic and social repercussions.

Future research should delve into the interrelations between the psychological and social determinants guiding homeowner decisions; nuances and linkages with the rental market; risks to broader financial markets; and the impact of other water-related hazards such as subsidence. Similarly, exploring homeowner behaviour via longitudinal studies—particularly regarding buying and selling decisions post-flood—would offer valuable insights. By advancing this knowledge, we pave the way toward more resilient communities that can deal with the multifaceted challenges posed by flooding.

Methods

Property transaction data and repeat-sale identification

We use residential property price transaction data from the UK Land Registry (UKLR) to demonstrate what can be achieved with publicly available data⁹⁰. The UKLR contains detailed information about residential property transactions spanning 1st January 1995 to present day, although recent months are often updated retrospectively, so can be partially incomplete. Our study period utilises data from the 28-year period 1st January 1995 to the 1st January 2023.

We apply a natural logarithm transformation to the price (£) of each transaction, determine the mean and standard deviation of this log price, and calculate a threshold based on four standard deviations from the mean of the log price. We then convert these log thresholds back to original price values and count the number of transactions lying outside these thresholds. We truncated the dataset by removing rows with prices outside the thresholds, effectively removing extreme outliers. Our upper and lower thresholds were £3,718,654.30 and £5,557.30. This leaves 27,758,154 recorded transactions (after 19,438 outliers were removed).

The price-paid dataset encompasses property transactions from both England and Wales. Therefore, we filter to exclude property transactions outside of England. This is done by comparing transaction records against all English postcodes as of January 2023.

Following this, the dataset was scrutinised for entries with incomplete or missing information key to identifying repeat sales. Fields such as

postcode, primary addressable object name (PAON), secondary addressable object name (SAON), and street name were examined for null or empty values. Additionally, transactions with unlikely attributes (e.g., sale prices at or below £100, or properties categorised under a type like 'Other' that might have been missed) were identified and excluded.

Finally, we generate repeat-sale pairings. This process selects all transactions from the cleansed dataset and augments them with a unique identifier for each property, by concatenating relevant fields (see supplementary material). Subsequent transactions for the same property were identified by comparing sale dates and prices, allowing for isolation of pairs of transactions related to the same property. After all this we were left with 12,009,213 sale pairings for analysis.

Postcode-district and local authority district data

For our postcode-district (the highest spatial resolution available) data, we used the Ordnance Survey Code-Point with Polygons dataset⁹¹. This was derived from the AddressBase, which is another Ordnance Survey dataset that provides National Grid reference coordinates for every postal delivery address in Great Britain.

Local Authority District (LAD) data were obtained from the Office for National Statistics⁹². Given the sometimes marked changes associated with local authority district boundaries due to historical changes in local government in England—and to avoid any issues related to mismatching postcodes with outdated local authority district boundaries—we updated the local authority district associated with each postcode-district to conform to the 2023 LAD dataset from the ONS. For more information, please refer to the supplementary material.

Flood extent data

To distinguish between properties affected by flooding (treatment group) and those unaffected (control group), the variable NFHU captured whether or not a flood had occurred in the same district as the property, within the sale pairing. Here, a district was represented by the LAD boundaries from the Office for National Statistics as of December 2023⁹². Our aggregation involved classifying sales transactions based on whether they 'bracket' a flood event within the same district as the property itself, thereby facilitating a comparison of price changes for properties directly affected by flooding. NFHU refers to a sale pairing that happened before and after a nearby flood event. This helped to identify how property values were impacted by flooding, by comparing their sale prices before and after such events.

Flood extent data for England are available from the Environment Agency as part of their Recorded Flood Outlines dataset—the best estimate of historical flood extents in the public domain⁹³. We used this information to calculate whether a sale pairing had experienced a flood event or not, and to compute flood history for each postcode district in England. This dataset includes recorded events from fluvial, coastal and sewer flooding. In total there were 78,025 polygon events covering an area of 2723 km². Fluvial flood events accounted for 89% (2414 km²), whereas coastal and sewer flood events covered 6% (170 km²) and 5% (139 km²) respectively.

We used the Environment Agency Flood Zone 3 dataset to calculate the proportion of a postcode area lying within the high-risk Flood Zone 3⁹⁴. The Flood Map for Planning (Rivers and Sea) is the best estimate of areas of land at risk of flooding, when the presence of flood defences is ignored. This covers land with a 1% or greater chance of flooding each year from rivers, or a 0.5% or greater chance of flooding each year from the sea.

Calculating flood history

First, we removed all recorded flood extents with start dates prior to 1st January 1995. Using the Environment Agency flood group identifier, we removed duplicate entries forming a unique flood lookup table. Then we grouped these separate flood events by postcode-district, forming a dataset where unique flood events were aggregated at the postcode-district scale giving us the number of flood events that a given postcode-district had experienced during the study period. We then performed quality control checks for instances where the end date of a flood event is before the start

date and vice versa. Upon review, some errors arose from American and UK datetime formatting mix-ups. Please refer to the supplementary material for a worked example.

Flood risk data

By using the postcode-district data alongside the Flood Zone 3 extent polygons, we calculate the area of each postcode in England, and then the proportion of each postcode covered by Flood Zone 3. This gives a value between 0 and 1 for Flood Zone 3 coverage for every postcode in England.

Hedonic price modelling

To estimate the impact of flooding on property prices, we employ a repeat-sales hedonic price model, adhering to best practices to control for time-invariant characteristics of properties. This method has been applied in other studies to estimate the effects of flood events on property prices, mortgages, and flood zonation^{4,8,16,95}. Hedonic price modelling assumes that a property price reflects the sum of its characteristics and attributes, including site-specific environmental factors such as flood exposure and historical flooding. By analysing changes in sales prices over time for properties sold at least twice within our study period, we isolate the effect of flood risk on property valuation. The hedonic model coefficients then reveal the relative weight attached to these factors in space and time. Here, the environmental factors considered were relative exposure to the risk of flooding and history of flooding for a given property. For a summary of all variables used and associated summary statistics, please refer to Tables S1 and S2 in the supplementary materials.

Age, period and cohort effects

To distinguish post-flood recovery dynamics from broader market and institutional changes, we extended the repeat-sales hedonic model using an age-period-cohort (APC) framework. Age effects capture the dynamic price path following a flood event and were represented by the post-flood recovery interval dummies. Period effects capture broader shocks affecting all properties, irrespective of their flood history, and were represented using dummy variables based on the second sale date: 1995–2008 (baseline), 2009–2015, and post-2016. These periods reflect major changes in the economic and flood insurance environment, including the 2008–09 Global Financial Crisis, the expiry of the Statement of Principles, and the introduction of Flood Re.

Cohort effects capture differences associated with the period in which a property's postcode first experienced recorded flooding. Cohort dummies were therefore defined using the first recorded flood date in each postcode: pre-2009, 2009–2015, and post-2016, with non-flooded properties as the baseline comparison. Failure to account for these distinct temporal processes can lead to both conceptual misinterpretation and statistical bias. For example, a prolonged price depression might be misattributed to slow market learning (an age effect) when in fact it reflects a one-off repricing following a policy shift (a period effect). Similarly, what appears to be a cohort-based penalty may disappear once broader period shocks are controlled for. For more information on the definitions behind the three time horizons, please refer to the supplementary material.

Survival analysis

To investigate the impact of flooding on length of home ownership, we utilised survival analysis (time-to-event) techniques—a branch of statistics commonly used for analysing duration-based data^{96–98}. Specifically, we utilise a Kaplan-Meier estimator as part of the open-source Python package 'lifelines' to analyse length of home ownership for various flood exposures⁹⁹ (Table 1). The Kaplan-Meier estimator is a non-parametric statistic used to estimate the survival function from duration-based data, allowing calculation of the probability that homeowners will remain in their property for a given length of time, having accounted for incomplete observations (i.e., homeowners remaining in the same property beyond the study period). For more details on how we calculate survival curves, please refer to the supplementary materials.

Data availability

All data used in this study are publicly available. Residential property transaction data were obtained from HM Land Registry's Price Paid Data⁹⁰. Recorded Flood Outlines and Flood Zone 3 extents were obtained from the Environment Agency's open data services^{93,94}. For further information on how these data were used, please refer to the supplementary material.

Code availability

All code used as part of this manuscript is available from the corresponding author on reasonable request. Privileged access to specialised code can create a disparity whereby those with access benefit disproportionately, potentially disadvantaging others. In fields such as real estate, where property values significantly influence individual wealth, the implications of such disparities can be profound. Our code, developed for the preceding study, could influence property valuations and market dynamics if misused. Therefore, to safeguard the financial interests and well-being of communities involved, the code is made available only upon request. Although the data used are publicly accessible, unrestricted dissemination of the code could lead to unintended negative consequences for homeowners and local markets. While we support transparency and the broader goal of advancing climate adaptation through research, we recognise that making the code publicly available might unfairly advantage certain groups and harm those in vulnerable positions.

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Author contributions

J.J.T. acquired and analysed the data, produced the figures, and conceived and implemented the experimental design and took lead in writing the paper. R.L.W. and J.K.H. contributed to the experimental design. R.L.W., J.K.H., R.C., and N.G. contributed to interpretation of the data, offered insightful critiques, and influenced the final content. All authors critically reviewed, edited, and contributed to the final paper.

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Competing interests

The authors declare no competing interests.

Additional information

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